

FRACTURE DETECTED

Confidence: 69.1%

Original Radiograph

AI Attention Map (Grad-CAM)

Clinical Findings

Classification: Oblique DisplacedICD-10: S42.33Severity: Moderate to Severe

Definition:

An angled fracture where bone fragments have shifted from normal alignment.

Simplified Explanation

The heatmap on your X-ray highlights areas of concern within your lower leg and ankle. The bright red and yellow areas indicate where the radiologist's attention was focused due to signs of an oblique displaced fracture. This means the bone has broken at an angle, and the broken pieces have shifted out of their normal alignment.

To properly manage this fracture, you will likely require further evaluation and treatment. This may include immobilization with a cast or splint, and potentially surgery to realign the bone fragments. Your doctor will discuss the best treatment plan for your specific situation, which will involve pain management, preventing further injury, and a rehabilitation program to help you regain full function of your leg and ankle.

Conformal Prediction Set

Statistically guaranteed inclusion set (90% coverage)

Model
Maxit
Hypercolumn Cbam Densenet169
Rad Dino
Yolo

Prediction
Oblique Displaced
Oblique Displaced
Oblique Displaced
Oblique

Ensemble Composition

Individual model predictions

Confidence
99.9%
99.4%
77.2%
99.7%

Detailed Clinical Analysis

Diagnosis: Oblique Displaced • Powered by Gemini

Clinical Analysis: Oblique Displaced Fracture

This patient presents with an oblique displaced fracture. The provided information indicates that this is an angled fracture where bone fragments have shifted from their normal alignment, classified as moderate to severe in severity.

Fracture Classification and Location:

The diagnosis of "Oblique Displaced" provides the fracture pattern. However, a specific bone and exact anatomical location are not provided in the current context. To precisely classify this fracture according to established systems like the AO/OTA (Arbeitsgemeinschaft für Osteosynthesefragen/Orthopaedic Trauma Association) classification, the affected bone (e.g., tibia, femur, humerus) and the specific segment (diaphyseal, metaphyseal, or epiphyseal) are critical. An oblique fracture is characterized by its oblique orientation relative to the long axis of the bone. The displacement signifies that the fracture fragments are not in their normal anatomical position.

ICD-10 Coding:

The provided ICD-10 code is S42.33. This code specifically refers to an "Oblique fracture of shaft of humerus, unspecified arm." This implies the fracture is located in the diaphysis of the humerus. If the fracture involves a different bone, the ICD-10 code would change accordingly. For example:

- * Oblique fracture of shaft of tibia: S82.23
- * Oblique fracture of shaft of femur: S72.33

It is crucial to verify the exact anatomical location to confirm the ICD-10 code's accuracy.

Evidence-Based Management Pathways:

The management of oblique displaced fractures is guided by fracture stability, degree of displacement, comminution, involvement of articular surfaces, and patient factors.

1. Conservative Management:

- * Indications: Non-displaced or minimally displaced oblique fractures with inherent stability may be managed conservatively.
- * Protocol: This typically involves closed reduction (if displacement exists) followed by immobilization in a cast or brace. The duration of immobilization is often extended, typically 8-12 weeks, to allow for adequate healing, especially in oblique fractures which can be prone to displacement due to the shearing forces acting along the oblique line. Regular radiographic monitoring is essential to assess alignment and detect any secondary displacement.
- * Limitations: Oblique fractures, by virtue of their angulation, are often less stable than transverse fractures and can be susceptible to displacement, particularly under gravitational or functional loads. Therefore, conservative management may be less successful for significantly displaced or unstable oblique fractures.

2. Surgical Management:

- * Indications: Significantly displaced oblique fractures, those with instability, open fractures, intra-articular extension, or failure of conservative management are typically indications for surgical intervention.
- * Surgical Options:
 - * Internal Fixation: This is the mainstay for displaced oblique fractures. The choice of implant depends on the bone and fracture location:
 - * Plates and Screws: A dynamic compression plate (DCP) or locking compression plate (LCP) applied to the tension side of the bone, often with interfragmentary lag screws placed across the oblique fracture line, can provide excellent stability. This method is particularly effective in achieving compression at the fracture site and resisting rotational and bending forces.
 - * Intramedullary Nailing (IMN): For diaphyseal fractures of long bones like the femur or tibia, antegrade or retrograde IM nailing can be an option, especially for long oblique patterns. However, for shorter oblique fractures, IM nailing alone might not provide sufficient rotational control, and may require adjuvant fixation.
 - * K-wires and Pins: In certain scenarios, particularly in pediatric fractures or for small fragments, K-wires might be used, often in conjunction with other fixation methods or for maintaining reduction.
 - * Open Reduction and Internal Fixation (ORIF): This is the standard approach when surgical intervention is required, allowing for direct visualization and anatomical reduction of the fracture fragments, followed by stable internal fixation.